

**Foundation Mathematics 1017SCG**  
**Week 5 Workshop Questions**

1. Solve the following equations.

(a)  $5x + 4 = 19$

(b)  $7x - 2 = 33$

(c)  $3x + 9 = -3$

(d)  $-2x + 6 = 26$

(e)  $\frac{x}{5} + 2 = 7$

(f)  $\frac{2x}{3} - 4 = 0$

(g)  $6n + 8 = 44$

(h)  $5(3m - 1) = 70$

(i)  $-18 - 3h = 5$

(j)  $4x + 7 = 8 - x$

(k)  $2(4y + 1) = 5(3 - y)$

(l)  $7(5 - c) = 3(c + 1) - 10$

2. Solve the following equations giving **exact** answers. An exact answer has NOT been rounded at any step in the working.

(a)  $x^2 + 5x = 0$

(b)  $x^2 - 4x = 0$

(c)  $x^2 - 9 = 0$

(d)  $x^2 - 13x + 42 = 0$

(e)  $3x^2 + x - 2 = 0$

(f)  $2x^2 = 5x + 3$

(g)  $2x^2 + 4x - 6 = 0$

(h)  $x^2 = 7x - 4$

(i)  $2x^2 - 9x + 5 = 0$

(j)  $5x^2 - 3x - 1 = 0$

3. Solve the following equations.

(a)  $\log_2(x) = 4$

(b)  $\log_5(x) = 3$

(c)  $6\log_4(x) = 12$

(d)  $\log_3(x) + 7 = 10$

(e)  $\log_{10}(5x) = 2$

(f)  $4\log_7(9x) = 8$

(g)  $8\log_2(4x) = 24$

(h)  $5\log_3(6x - 1) = 10$

(i)  $9\log_5(3x + 2) + 7 = 25$

(j)  $3\log_4(5x + 8) - 4 = 2$

4. Solve the following equations. Where necessary, round your answer to two decimal places.

(a)  $2^x = 12$

(b)  $5^x = 30$

(c)  $9^x = 58$

(d)  $e^x = 4$

(e)  $2e^x = 12$

(f)  $7e^x = 21$

(g)  $5e^x + 4 = 14$

(h)  $3e^{2x+1} = 15$

(i)  $8e^{5x-4} = 32$

(j)  $6e^{9x+7} + 2 = 20$

5. Solve the following equations. Remember that when adding or subtracting fractions, a common denominator is needed.

(a)  $\frac{x}{3} + \frac{4x}{5} = \frac{1}{3}$

(b)  $\frac{15y}{4} - 5 = \frac{y}{2}$

(c)  $\frac{2x-4}{5} + 6 = \frac{x}{2}$

(d)  $\frac{4h-1}{2} - \frac{2h+5}{3} = 0$